

ORIGINAL ARTICLE

Morphology of four common and phylogenetically distant ant species varies along disturbance and aridity gradients in the Caatinga dry forest

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Abstract

Understanding species adaptations to human-modified ecosystems is central to evolutionary and conservation biology. It is essential to evaluate trait variations in populations that occur along environmental gradients to understand the adaptive potential of species to human-driven environmental change. We analyzed morphological trait variations in the workers of four common and phylogenetically distant ant species (*Dinoponera quadriceps*, *Dorymyrmex goeldii*, *Pheidole radoszkowskii*, and *Ectatomma muticum*) along disturbance and aridity gradients in the Caatinga dry forest. Ants were collected using pitfall traps in 20 0.1-ha plots distributed over an area of 213.75 km². Up to five individuals per species were selected from each plot to measure eight morphological traits. Between-plot variability contributed considerably (~37%) to the total variability of morphological traits. This pattern of high between-plot variability was consistent across species and traits. Such morphological intraspecific variations between plots were not strongly related to the aridity or disturbance gradients. However, when ant morphology did respond to environmental change, the response was stronger to aridity rather than disturbance, and responses were most conspicuous in *P. radoszkowskii*. Our results suggest that ants in the Caatinga may have substantial potential for morphological adaptation to environmental changes and that disturbance intensity and variations in aridity are not strong determinants of ant worker morphology despite influencing morphological variation over small geographical distances. These morphological variations are likely to represent key assets, by enabling individuals and population to inhabit the variety of habitats provided by Caatinga and to cope with upcoming changes in land use and climate.

Abstract in Portuguese is available with online material.

KEYWORDS

acclimation, adaptation, chronic anthropogenic disturbance, climate change, environmental filtering, functional traits, rainfall, semi-arid tropics

1 | INTRODUCTION

Understanding species adaptations to ecosystems altered by human activity is fundamental to evolutionary and conservation biology. For species to persist in remaining disturbed habitats, they must have suitable physiological, morphological, behavioral, and life history functional traits for the environmental filters imposed by human impacts (Southwood, 1988; Violle et al., 2007); have plasticity in the expression of their traits (Beever et al., 2017; Hadfield & Strathmann, 1996); or adapt to new environmental conditions through natural selection (Hadfield & Strathmann, 1996). Morphological traits are widely used to assess species responses in relation to environmental gradients (Arnan et al., 2012; Gibb & Parr, 2013; Pigot et al., 2016; Sfair et al., 2018). Since body size is strongly related to organism physiology (Peters, 1983) and resource exploitation (Okuzaki et al. 2009, Stouffer et al., 2011), body size is the most used morphological measure to assess species' responses to environmental gradients. For instance, according to Bergmann's rule, individuals from the same species tend to be larger in cooler environments (Bergmann, 1847; Blackburn et al., 1999; Weeks et al., 2020). Other morphological traits can also be important for explaining species adaptations to environmental change. For instance, the size-grain hypothesis (Kaspari & Weiser, 1999) states that species with smaller legs can take advantage of more complex environments, making it easier to move between cracks, and such effects might also translate to the population level (Chown & Nicolson, 2004).

Ants are a suitable model system to analyzing morphological variation along environmental gradients because they exhibit high levels of morphological and functional variation, in addition to being extremely sensitive to environmental change (Andersen & Majer, 2004; Underwood & Fisher, 2006), especially in the tropics (Hölldobler & Wilson, 1990). During the last decade, a large and growing literature of morphological trait-based analyses on worker ants along environmental gradients has been published (e.g. Bihn et al., 2010; Gibb et al., 2018; Parr et al., 2017; Silva & Brandão, 2010; Yates et al., 2014) including studies linking morphological traits and ecological functions (Bihn et al., 2010; Kaspari & Weiser, 1999; Parr et al., 2017). These studies focus on average species trait values and assume that intraspecific variation is small or negligible. Thus, we still know little about morphological changes at the population level in ants. A recent study found very low levels of intraspecific variation along an elevation gradient (1–4% of total trait variation), probably masked by the large variation among species that accounted for most of the total variation (Gaudard et al., 2019). Moreover, Gibb et al. (2018) suggest that recent evolutionary pressures on ants can generate many relationships between morphological traits and the environment. Furthermore, recent studies have demonstrated that workers of *Leptothorax acervorum* and *Aphaenogaster senilis* populations are larger at higher altitudes (Bernadou et al., 2016; Shik et al., 2019).

Here, we analyze the variation in eight morphological traits of four common and phylogenetically distant ant species, and test whether this variation responds to disturbance and aridity gradients in the Caatinga dry forest. The Caatinga is one of the world's most highly populated and threatened semi-arid biomes, sustaining

28 million people who are highly dependent on forest natural resources for their livelihoods in a typical regime of chronic anthropogenic disturbance (Silva et al., 2017). In addition, Caatinga is highly threatened by climate change, with climate models forecasting a 22–40% reduction in rainfall and a 3–6°C increase in temperatures by 2100 (Magrin et al., 2014). Increasing aridity has been observed even within small geographic ranges (Rito et al., 2017). Thus, Caatinga offers an excellent opportunity to study potential morphological adaptations to disturbance and climate change. Moreover, previous research on the Caatinga has shown that chronic anthropogenic disturbance and/or aridity do not affect ant abundance or species richness, but does alter species composition (Arnan, Arcoverde, et al., 2018; Câmara et al., 2018; Oliveira et al., 2017) and spatial-temporal foraging activity of widely distributed species (Silva et al., 2019). Despite this, many species are unaffected and occur along a large portion of the disturbance and/or aridity gradients. Morphological traits may play an important role in how ant populations respond to increases in chronic disturbance and aridity gradients.

Our specific aims are: (1) to characterize and quantify the levels of morphological intraspecific variation within and between study plots throughout the region, (2) to verify to what extent morphological intraspecific variation depends on species or morphological trait identity, and (3) to test to what extent morphological intraspecific variation responds to changes in disturbance and aridity. We predict that: (1) the degree of morphological intraspecific variation will depend on species or morphological trait identity, as previous studies have suggested that some ant clades exhibit greater trait conservatism than others (Hölldobler & Wilson, 1990; Silva & Brandão, 2010; Weiser & Kaspari, 2006), and some traits tend to be more plastic than others (Gaudard et al., 2019; Oms et al., 2017; Purcell et al., 2016; Yates et al., 2014); and (2) morphological intraspecific variation will respond to changes in disturbance and aridity, as they are the strongest environmental filters in the study area (Arnan, Arcoverde, et al., 2018; Rito et al., 2017).

2 | MATERIAL AND METHODS

2.1 | Study area

The study was carried out in Catimbau National Park (8°24' 00"–8°36' 35"S; 37°09' 30"–37°14' 40"W), a 607-km² protected area located in the state of Pernambuco, northeastern Brazil (Figure S1). The climate is semi-arid, with annual temperature averaging 23°C, and mean annual rainfall of 600 mm, concentrated between March and July. Approximately 70% of the park area is covered by quartzite sandy soils, but planosols (15%) and lithosols (15%) are also present. The vegetation is composed of shrubs and small trees up to 7 m in height and is dominated by the Leguminosae, Euphorbiaceae, and Myrtaceae families (Rito et al., 2017). Despite the park being established in 2002, local inhabitants still remain within its borders; therefore, it continues to be subject to a regime of chronic anthropogenic disturbance, including livestock grazing, timber harvesting, firewood collection, and

hunting (Arnan, Leal, et al., 2018; Rito et al., 2017). Catimbau includes a very steep rainfall gradient from 1100 mm in the south-east to 480 mm in the northwest (Rito et al., 2017); such bioclimatic gradients provide powerful frameworks for addressing climate change impacts through space-for-time substitutions (Blois et al., 2013; Caddy-Retalic et al., 2017). We sampled twenty permanent plots (Figure S1) measuring 20 m × 50 m (0.1 ha) that were established for several studies carried out as part of the Long-Term Ecological Research Project PELD-Catimbau (www.peldcatimbau.org); they are separated by at least 2 km within a total area of 213.75 km² (Rito et al., 2017).

2.2 | Characterizing chronic anthropogenic disturbance and aridity gradients

To assess the effects of disturbance on the variability of ant morphology, we used a global multi-metric disturbance index previously established for the study plots (Arnan, Leal, et al., 2018). We integrated into a single index nine disturbance metrics that are related to the three main sources of chronic anthropogenic disturbance in the Caatinga: livestock grazing, wood extraction, and exploitation of non-timber forest products (Arnan, Leal, et al., 2018; Rito et al., 2017). For livestock grazing, we conducted field measurements of goat-trail length and density of goat and cattle dung. For wood extraction, we measured biomass of live stems and coarse woody debris in the field. For exploitation of non-timber products, we used four surrogates of proximate human pressure: proximity to the nearest house, proximity to the nearest village, proximity to the nearest road (using satellite imagery in ArcGIS 10.1 software), and number of people in the nearest village (obtained by interviewing local households) with influence weighted by distance. The values for each disturbance metric were first standardized between 0 and 1 to make component metrics of equal importance. We then computed the global multi-metric index by summing the three components and multiplying by 100. The overall index can potentially vary from 0 (zero values for all component metrics) to 100 (maximum values of all component disturbance metrics). However, it ranged from 2 to 58 in our study area. Full details on this disturbance index are provided in Arnan, Leal, et al. (2018).

We obtained water deficit/aridity data from hydrological maps of the CGIAR-CSI's Global Aridity and PET Database and Global High-Resolution Soil Water Database (Trabucco & Zomer, 2010). These maps have a 30-sec resolution, equivalent to 1 km. To plot these data onto maps, the annual average water deficit was calculated from the difference between the mean annual potential evapotranspiration (PET) and the actual evapotranspiration (AET) using the ArcGIS 10.1 software. Water deficit ranged between 658 mm and 1086 mm.

2.3 | Ant sampling

In each of the 20 plots, we established 20 pitfall traps (in a 4 × 5 grid) positioned 5 m apart. Traps were constructed from 12-cm-diameter plastic pots partially filled with an alcohol and ethylene glycol solution with detergent to preserve the sample and

were activated for 48 h in March 2015 (for more details on ant sampling, see Arnan, Arcoverde, et al., 2018). We selected the four most widely distributed species (i.e., species occurring in ≥10 plots) in our study landscape to measure morphological attributes, which belonged to four different subfamilies following an evolutionary gradient from the closest to the baseline to the most evolutionary recent species (*Dinoponera quadricaps* (Ponerinae), *Dorymyrmex goeldii* (Dolichoderinae), *Pheidole radoszkowskii* (Myrmicinae), and *Ectatomma muticum* (Ectatomminae)) (Moreau & Bell, 2013). These species also follow a gradient of worker size, with *D. quadricaps* displaying the largest workers, *P. radoszkowskii* the smallest workers, and *E. muticum* and *D. goeldii* exhibiting intermediate sized workers. For each species and plot, we measured between 3 and 10 (median = 5) randomly selected workers from the captured individuals. A recent study showed that, on average, measuring around six individual worker ants is enough to generate a trait mean close to the population mean (Gaudard et al., 2019). Moreover, many studies and databases of ant traits use between three and 10 individuals (Bishop et al., 2015; Gibb et al., 2014; Parr et al., 2017; Silva & Brandão, 2010). Out of the 20 plots, we could only collect between 3 and 5 individuals in 14 plots for *D. quadricaps*, 13 plots for *E. muticum* and *P. radoszkowskii*, and 10 plots for *D. goeldii*. The plots where each species occurred were well distributed along both our chronic anthropogenic disturbance and aridity gradients, except for *Dorymyrmex goeldii* that only occurred in the most arid plots (Figure S2).

2.4 | Morphological traits

For each selected individual, we measured eight morphological traits that may respond to environmental changes and that are particularly sensitive to variations in aridity and chronic anthropogenic disturbances (Table 1). For each individual, we measured body size (Weber's length), eye length, antenna (scape + funicle) length, mandible length, clypeus length, leg (tibia + femur) length, inter-ocular distance, and pilosity (Table 1). These traits reflect body size, foraging capacity, foraging period, and resource acquisition strategy and can respond to environmental change (Bihn et al., 2010; Parr et al., 2017). We standardized all trait measurements (except Weber's length and pilosity) by dividing each by the Weber's length to limit correlations with body size. We then analyzed Weber's length, eye length ratio, antenna length ratio, mandible length ratio, clypeus length ratio, leg length ratio, inter-ocular distance ratio, and pilosity. *Pheidole radoszkowskii* possesses minor and major worker castes; only minor workers were measured. Pilosity was not measured for *Dorymyrmex goeldii* due to the absence of hairs in this species. Overall, 1934 trait measurements were taken.

2.5 | Statistical analyses

To understand the degree and source of variability in ant morphological traits, we performed one linear model per morphological trait

and species, where trait value was the response variable and plot was the fixed factor. Then, we computed the *between-plot* variability based on the Sum of squares (Sum Sq) of the plot factor and of the residuals from the model as follows:

$$\%_{\text{between}} = [\text{Sum Sq Plot} / (\text{Sum Sq Plot} + \text{Sum Sq Residuals})] * 100.$$

The *within-plot* variability was computed as: $\%_{\text{within}} = 100 - \%_{\text{between}}$. We then analyzed differences in the variability between plots among species and traits by means of a two-way ANOVA.

To characterize the overall variability in ant morphology between plots along the disturbance and aridity gradients, we first used a multivariate approach, performing a principal component analysis (PCA) of ant morphological traits for each ant species. All ant morphological traits were standardized prior to the analysis. Environmental variables (aridity and disturbance) were also plotted as supplementary variables in bi-dimensional PCA space (Legendre and Legendre, 1998). We then computed the mean and

TABLE 1 Morphometric traits used to characterize the morphological variation of workers of *Dinoponera quadricipes*, *Dorymyrmex goeldii*, *Pheidole radoskowskii*, and *Ectatomma muticum* along our disturbance and aridity gradients in Catimbau National Park, Pernambuco state, NE Brazil

Morphological trait	Measure	Function significance
Mesosoma or Weber's length	It represents body size and was measured from the base of the anterior slope of the pronotum to the lower posterior angle of the propodeum (Brown, 1953)	Strongly correlated with many physiological, ecological, and life history traits (Kaspari & Weiser, 1999)
Clypeus length ratio	Ratio of clypeus length to mesosoma length	Correlated with sucking ability and liquid-feeding behavior (Davidson et al., 2004)
Mandible length ratio	Ratio of mandible length to mesosoma length	Possible indicator of predatory lifestyle and thus types of resources consumed (Weiser & Kaspari, 2006)
Antenna length ratio	Ratio of antenna length (combined funicle and scape) to mesosoma length	Possibly correlated with ability to receive chemosensory information. Ants with long scapes may be more sensitive to pheromone trails (Weiser & Kaspari, 2006)
Leg length ratio	Ratio of leg length (combined length of femur and tibia) to mesosoma length	Possibly correlated with resource acquisition mode and foraging efficiency, as well as with the ability to cope with high foraging surface temperatures (Bihn et al., 2010; Kaspari & Weiser, 1999)
Eye length ratio	Ratio of eye length to mesosoma length	Likely correlated with main foraging period (day vs. night) (Bihn et al., 2010)
Eye position ratio	Ratio of inter-ocular width to mesosoma length	Related to hunting method (Fowler et al., 1991) or the component of the habitat occupied (Gibb & Parr, 2013)
Pilosity	Count of hairs crossing mesosoma profile	Hairs may increase tolerance to dehydration or may relate to mechanoreception (Wittlinger et al., 2007)

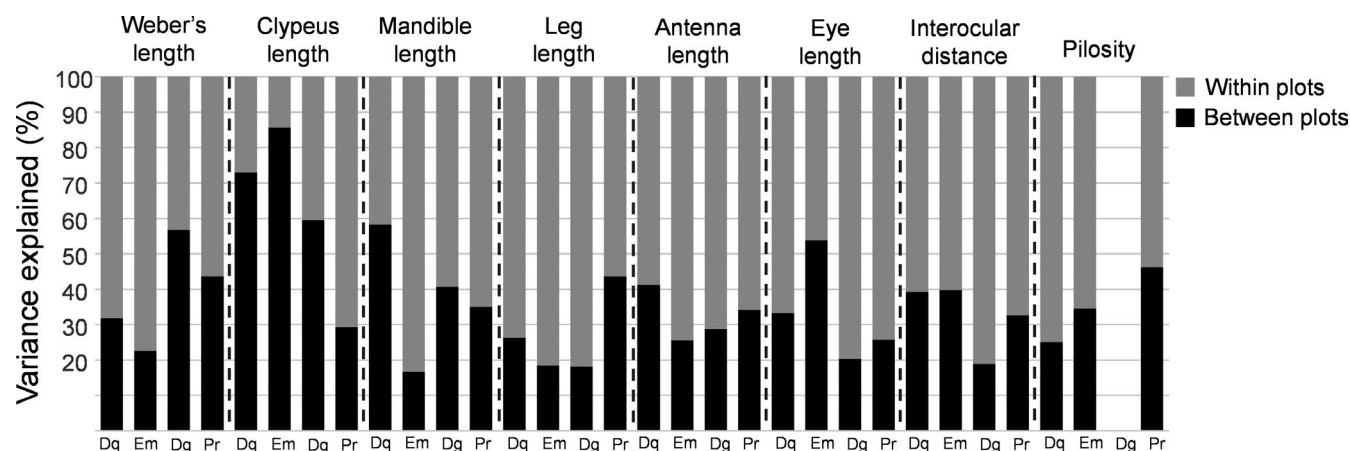


FIGURE 1 The within- and between-plot variance for each morphological trait and ant species in the Catimbau National Park, northeastern Brazil. Abbreviations: Dq, *Dinoponera quadricipes*; Em, *Ectatomma muticum*; Dg, *Dorymyrmex goeldii*; and Pr, *Pheidole radoskowskii*

the variance for each morphological trait, species, and plot, and constructed statistical models to test the effect of disturbance and aridity on variability in ant morphology (mean and variance) for each trait and species separately. We employed Akaike's information criterion with a correction for finite sample sizes (AICc) to select the best-supported models. This approach reduces the problems associated with multiple testing, co-linearity of explanatory variables, and small sample sizes (Burnham and Anderson, 2002). The best-supported models were selected based on their

AICc weights, which reveal the relative likelihood of a given model based on the data and the fit-scaled to one; thus, models with a delta (AICc difference) <2 were selected (Burnham and Anderson, 2002). The relevant variables were those that were retained in the best-supported models (except when the best-supported model consisted only of the intercept). Model selection was carried out using the dredge function in the MuMIn package for R (Barton, 2009). All computations and statistical analyses were performed in R (R Core Team, 2016).

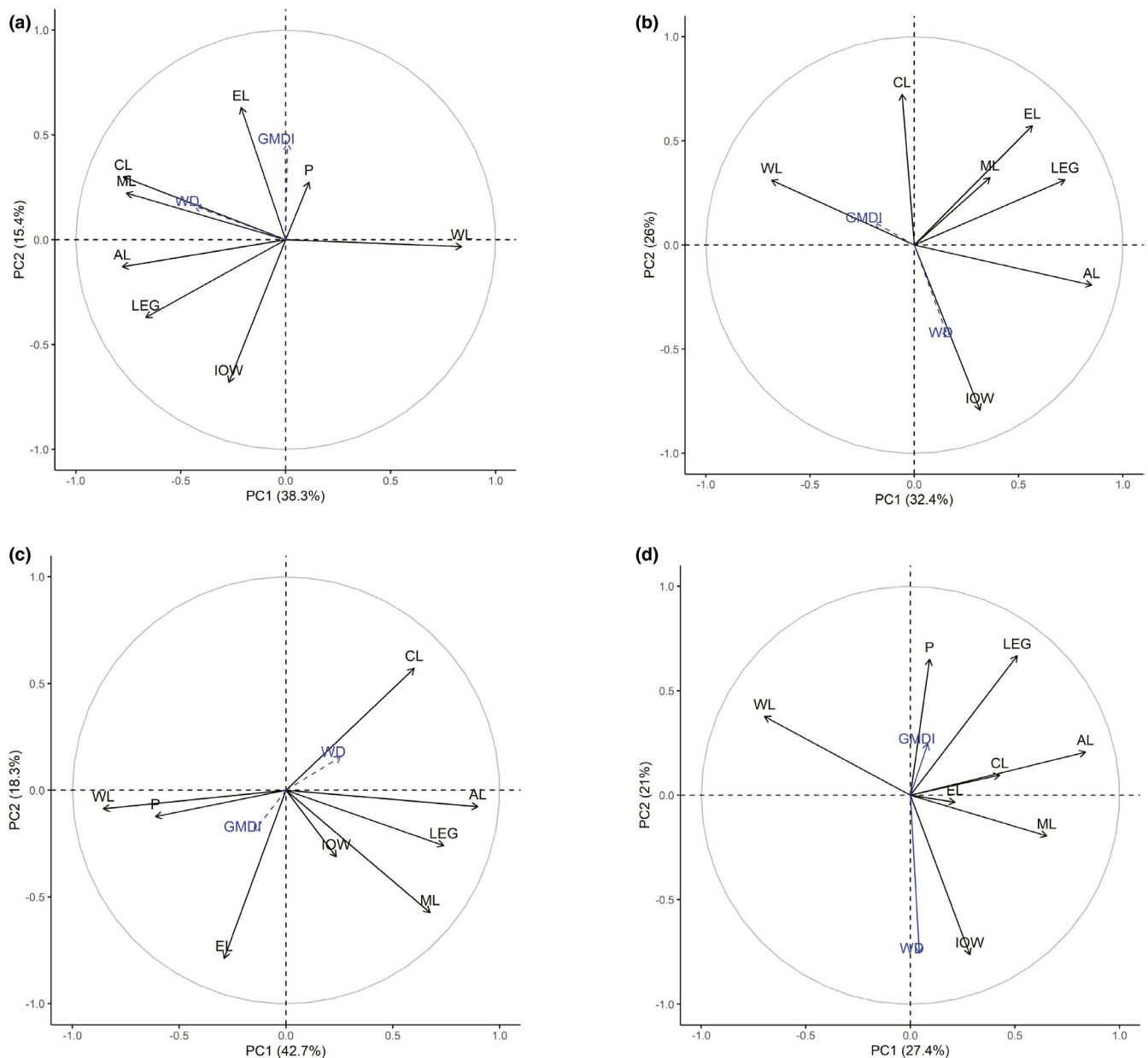


FIGURE 2 Principal component analysis (PCA) of ant morphological traits (black vectors) along aridity and disturbance gradients in the Catimbau National Park, northeastern Brazil, for four ant species: a) *Dinoponera quadriceps*, b) *Dorymyrmex goeldii*, c) *Ectatomma muticum*, and d) *Pheidole radoskowskii*. Aridity (WD) and disturbance (GMDI) are also plotted as supplementary variables (blue vectors) following Legendre and Legendre (1998). CL = ratio of clypeus length to mesosoma length, ML = ratio of mandible length to mesosoma length, WL = mesosoma or Weber's length, IQW = ratio of inter-ocular width to mesosoma length, EL = ratio of eye length to mesosoma length, P = count of hairs crossing mesosoma profile, LEG = ratio of leg length (combined length of femur and tibia) to mesosoma length, and AL = ratio of antenna (combined funicle and scape) length to mesosoma length

3 | RESULTS

Intraspecific variability within plots made the largest contribution to total variation for most morphological traits and species, but intraspecific variation between plots was considerable (Figure 1). The mean ($\pm$ SE) contribution of between-plot variability to total variability was $37 \pm 3\%$ (range: 16.7–85.6%) and did not differ among species (ANOVA; $F_{3,27} = 0.2$, $p = 0.861$) or morphological traits (ANOVA; $F_{7,23} = 1.9$, $p = 0.130$). The lowest values of between-plot variability ($<20\%$) were those of the mandible length ratio in *E. muticum*, the leg length ratio in *E. muticum* and *D. goeldii*, and the inter-ocular distance in *D. goeldii* (Figure 1). The highest values of between-plot variability ($>70\%$) were for clypeus length ratio in *E. muticum* and *D. quadriceps* (Figure 1). In general, most traits were not correlated with each other, and when they were, most were not highly correlated (i.e., $r < 0.70$) (Figure S3). Out of the 28 possible correlation coefficients for *D. quadriceps* and *E. muticum*, 10 were significant for *D. quadriceps* and 15 for *E. muticum*. However, only antenna length ratio and Weber's length were highly correlated ($r > 0.70$) in both species. *P. radoskowskii* was the ant species with fewest traits significantly correlated with each other (6 out of 28), and none of the correlation coefficients were greater than 0.70. For *D. goeldii*, out of the 21 possible correlation coefficients, six were significant and none of them were greater than 0.70 (Figure S3).

Overall, our multivariate analyses revealed that environmental variation is driving intraspecific variation among plots, especially for aridity. For species-level between-plot variability in ant morphology,

the first two PCA axes accounted for 53.7%, 58.4%, 61%, and 48.4% of the total variation in morphological data for *D. quadriceps*, *D. goeldii*, *E. muticum*, and *P. radoskowskii*, respectively (Figure 2). For *D. quadriceps*, aridity was more closely related to PCA1, and disturbance to PCA2 ($r = -0.43$ and 0.46 , respectively), suggesting that the traits most closely associated with PCA1 (Weber's length, antenna length ratio, clypeus length ratio, and mandible length ratio) were also more closely related to the aridity gradient, while the traits more closely related to PCA2 (inter-ocular width and eye length ratio) were more closely related to the disturbance gradient (Figure 2a). For *D. goeldii*, disturbance was weakly associated with both PCA axes ($r < 0.20$), and aridity was associated with PCA2 ($r = -0.44$) and, consequently, with the traits associated with this axis (inter-ocular width, clypeus length ratio, and eye length ratio) (Figure 2b). In relation to *E. muticum*, both aridity and disturbance were weakly correlated with both PCA axes ($r < 0.20$; Figure 2c). Finally, for *P. radoskowskii*, disturbance was weakly related to both PCA axes ($r < 0.20$), and aridity was strongly related to PCA2 ($r = 0.75$), suggesting that the traits most closely associated with PCA2 (leg length ratio and pilosity) were also most closely related to aridity (Figure 2d).

In agreement with our multivariate analyses, the species-level model selection analyses for each trait showed that most of the best-supported models retained aridity (Table 2). In general, *P. radoskowskii* was the species with the highest morphological variability between plots in response to the environmental gradients, followed by *D. quadriceps*, *D. goeldii*, and *E. muticum* (Table 2). The

TABLE 2 Variables retained in the best-supported models (delta AICc < 2) analyzing how ant morphological traits (mean and variance) for each species and each trait individually change along aridity (water deficit; WD) and disturbance (global multimetric disturbance index; GMDI) gradients in Catimbau National Park, Pernambuco state, NE Brazil

Ant species	Trait	Response variables	Explanatory variables	AICc	Delta AICc	Weight
<i>Dinoponera quadriceps</i>	CL	Mean	WD	40.4	0	0.635
	LEG	Mean	GMDI+WD	36.8	0	0.713
	EL	Mean	WD	37.2	0	0.62
	EL	Mean	GMDI+WD	38.4	1.14	0.35
<i>Dorymyrmex goeldii</i>	LEG	Mean	GMDI	30.9	0	0.613
	ML	Variance	WD	27.4	0	0.887
	AL	Variance	WD	29.2	0	0.813
<i>Ectatomma muticum</i>	P	Variance	GMDI	37.6	0	0.654
<i>Pheidole radoskowskii</i>	LEG	Mean	WD	30	0	0.875
	P	Mean	WD	36.8	0	0.78
	ML	Variance	GMDI	32.3	0	0.883
	LEG	Variance	WD	38.4	0	0.575
	LEG	Variance	GMDI+WD	40.1	1.72	0.244
	AL	Variance	GMDI	37.8	0	0.645
	P	Variance	GMDI	36.9	0	0.663

Note: The traits in which the best-supported model consisted only of the intercept were not included in the table. CL = ratio of clypeus length to mesosoma length, ML = ratio of mandible length to mesosoma length, EL = ratio of eye length to mesosoma length, P = count of hairs crossing mesosoma profile, LEG = ratio of leg length (combined length of femur and tibia) to mesosoma length and AL = ratio of antenna (combined funicle and scape) length to mesosoma length.

mean leg length ratio and pilosity of *P. radoskowskii* decreased with increasing aridity (Figure 3a,d); this species also showed greater variance in antenna and mandible length ratio, and pilosity with increasing disturbance (Figure 3b,c,f), and lower variance in leg length ratio with increasing disturbance and aridity (Figure 3e,g). *Dinoponera quadricaps* had a negative relationship between mean leg length ratio and disturbance intensity (Figure 4b) and a positive relationship between eye length ratio and disturbance intensity (Figure 4d); this species also showed a positive relationship

between mean clypeus length ratio, leg length ratio, and eye length ratio and aridity (Figure 4a,c,e). However, the variance of none of the traits of *Dinoponera quadricaps* was related to disturbance or aridity. For *D. goeldii*, the mean leg length ratio was negatively related to disturbance intensity (Figure 5b), and the variance in mandible length ratio decreased with increasing aridity, while the variance in antenna length ratio increased with increasing aridity (Figure 5a,c). Finally, in *E. muticum*, only the variance in pilosity responded negatively to increase in disturbance (Figure 6).

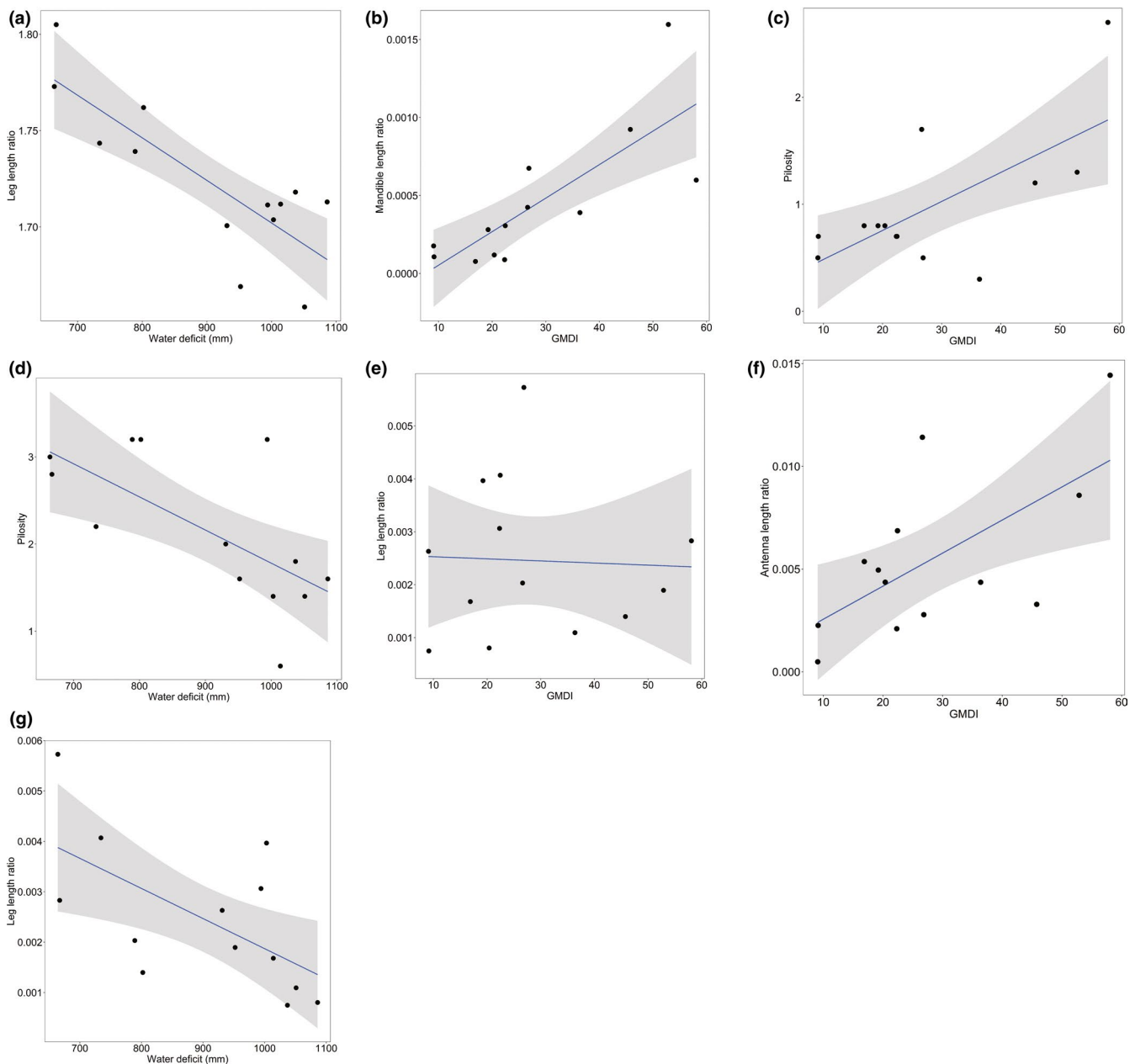


FIGURE 3 Linear relationships between ant morphological traits of *Pheidole radoskowskii* and the explanatory variables retained in the best-supported models in the Catimbau National Park, northeastern Brazil: (a) mean leg length ratio and water deficit; (b) variance in mandible length ratio and disturbance; (c) variance in pilosity and disturbance; (d) mean pilosity and water deficit; (e) variance in leg length ratio and disturbance; (f) variance in antenna length ratio and disturbance; and (g) variance in leg length ratio and disturbance

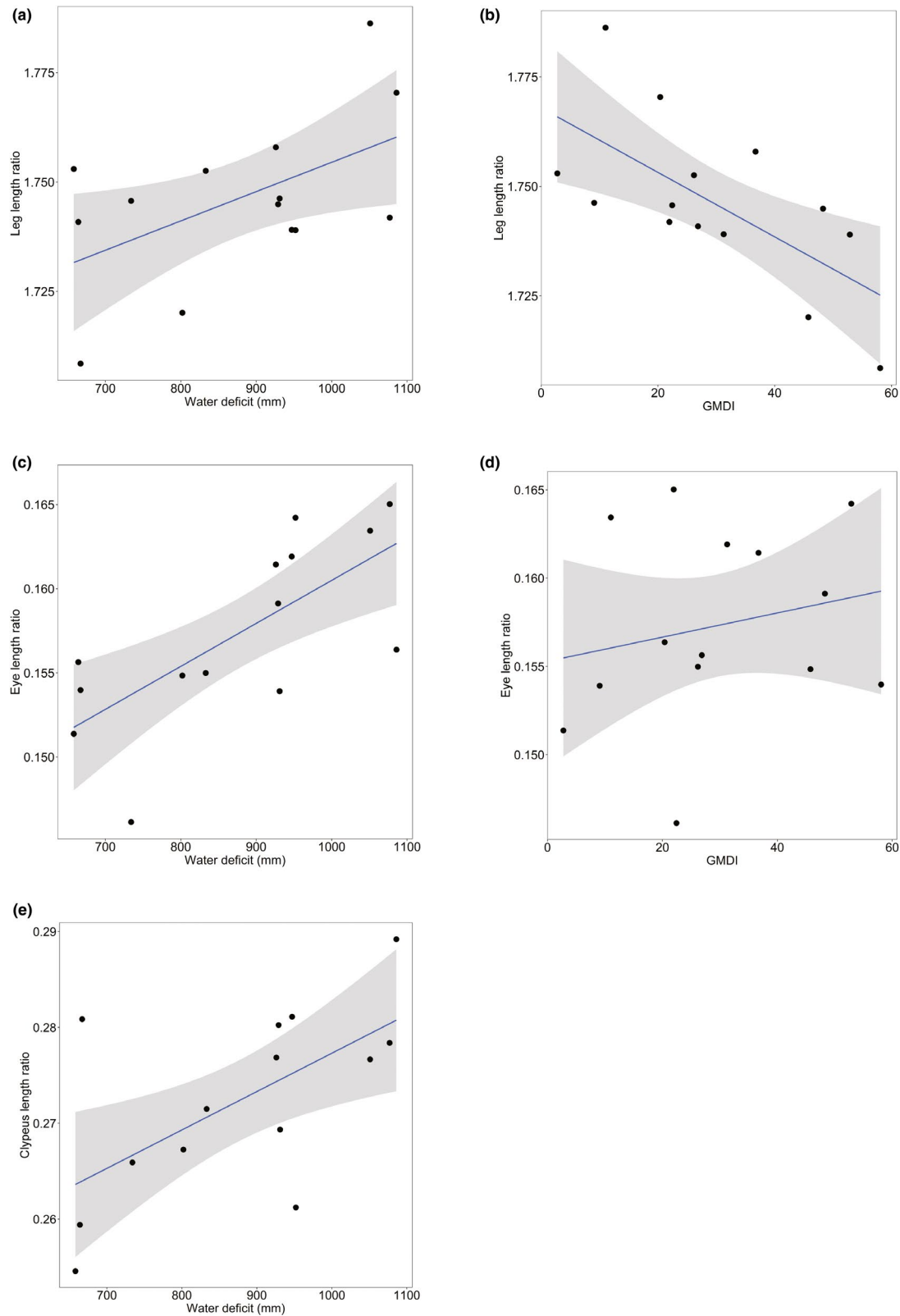


FIGURE 4 Linear relationships between ant morphological traits of *Dinoponera quadriceps* and the explanatory variables retained in the best-supported models in the Catimbau National Park, northeastern Brazil: (a) mean leg length ratio and water deficit; (b) mean leg length ratio and disturbance; (c) mean eye length ratio and water deficit; (d) mean eye length ratio and disturbance; and (e) mean clypeus length ratio and water deficit

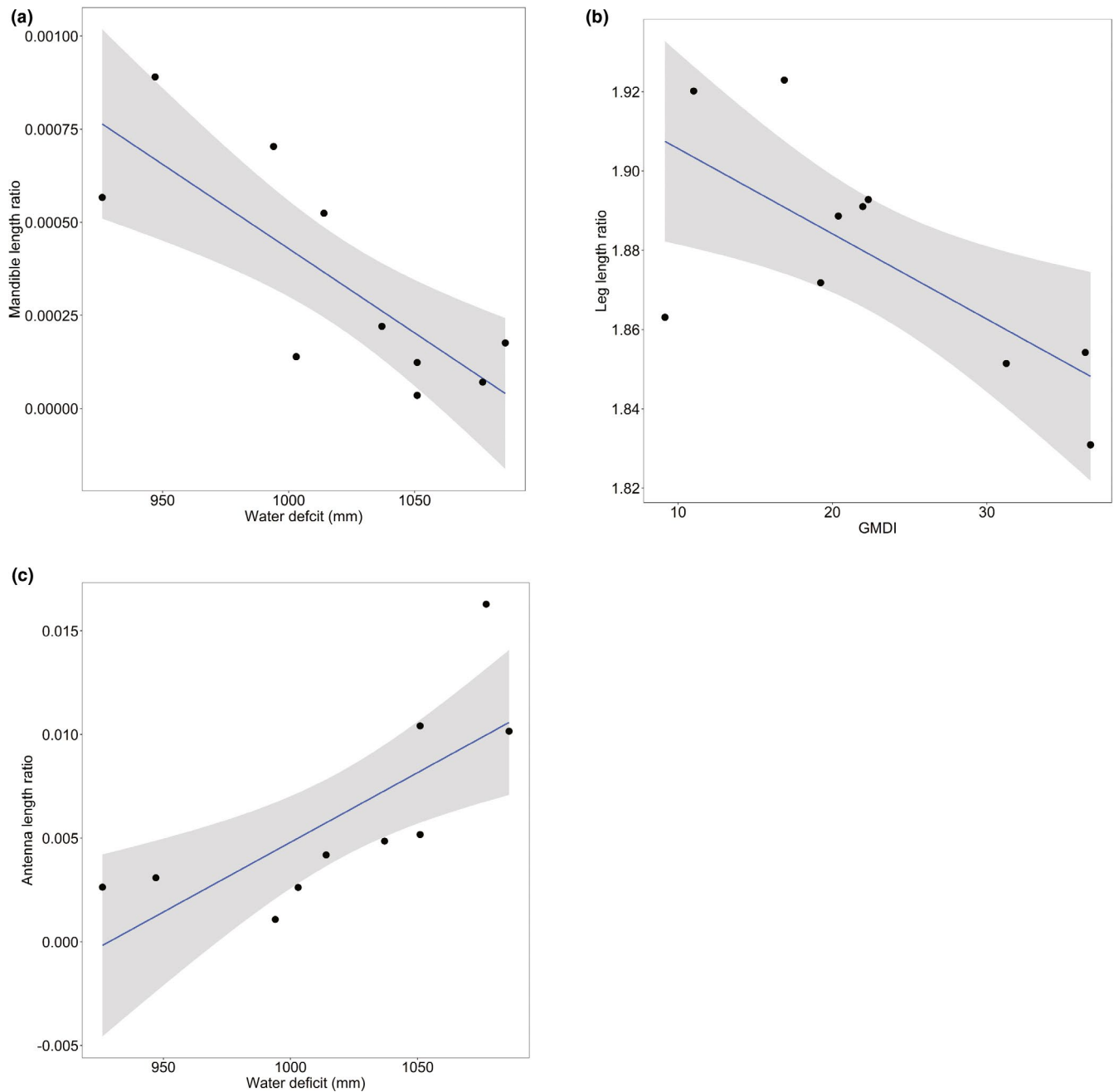


FIGURE 5 Linear relationships between ant morphological traits of *Dorymyrmex goeldii* and the explanatory variables retained in the best-supported models in the Catimbau National Park, northeastern Brazil: (a) variance in mandible length ratio and water deficit; (b) mean leg length ratio and disturbance; and (c) variance in antenna length ratio and water deficit

When assessing trait-level responses across all species, variability in leg length ratio was more closely linked to the environmental gradients than were the other traits (Table 2). Five out of the 15 best-supported models included leg length ratio as a response variable. Two out of the 15 best-supported models included mandible length ratio and eye length ratio, antenna length ratio, or pilosity as response variables. Only one of the best-supported models included clypeus length ratio.

4 | DISCUSSION

In this study, we report important morphological variation in the populations of four phylogenetically distant ant species over a small spatial (geographic) area in the Caatinga. Morphological variation is similar across ant species and traits and does not seem to be strongly related to either disturbance or aridity. However, the proportion of morphological variation that is related to environmental change is

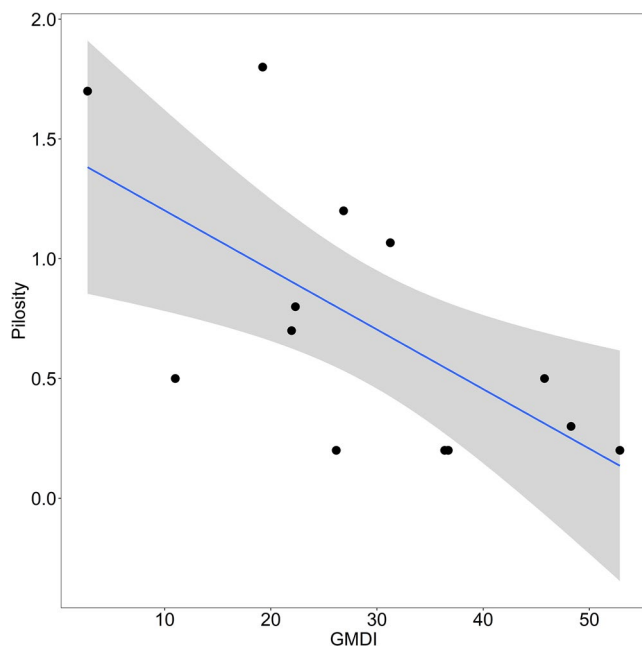


FIGURE 6 Linear relationship between variance in pilosity of *Ectatomma muticum* and disturbance in the Catimbau National Park, northeastern Brazil

most strongly associated with changes in aridity, especially in populations of *Pheidole radoskowskii* and a subset of morphological traits we studied. These results indicate the adaptation potential of ants for dealing with future changes in land use and climate (e.g., increased aridity).

The contribution of between-plot variability (~37%) to total morphological variability found here is in line with several studies on plants (Jung et al., 2010; Kazakou et al., 2014; Messier et al., 2010; Sfair et al., 2018), but differs to those reported for invertebrates, in which they display low levels of intraspecific trait variation, as found for dung beetles (3–5%, Griffiths et al., 2016) and ants (1–4%, Gaudard et al., 2019). However, while these studies analyzed a large number of species, we assessed only four ant species. Perhaps the use of the most widely distributed ant species along the gradients contributed to high between-plot intraspecific variation, as previously demonstrated for plants (Sfair et al., 2018). Species presenting a wide distribution must cope with a greater range of environmental constraints (Sanaphre-Villanueva et al., 2016; Sfair et al., 2018). Moreover, our four species may be considered generalists (Hölldobler & Wilson, 1990; Leal et al., 2017) and the similar high morphological variation shared by these species may be a mechanism that allows them to maintain populations in different environmental conditions.

Previous studies have shown that morphology (Silva & Brandão, 2010; Weiser & Kaspari, 2006) and other life history traits (Hölldobler & Wilson, 1990) are often conserved across sub-families and genera. However, more recent studies demonstrate that most analyzed ant morphological traits are phylogenetically

independent (Gaudard et al., 2019; Yates et al., 2014). Nevertheless, since we only analyzed four species, we cannot draw strong conclusions regarding evolutionary causes for the similar levels of morphological variation across the species and traits we observed.

Stronger effects of aridity compared to disturbance have previously been shown for dry forests, and this is likely because water is the main limiting factor in these ecosystems (Maass, 1995). Previous studies in the Caatinga dry forest have reported a similar pattern, albeit with other diversity measures such as the taxonomic diversity of plants (Rito et al., 2017), functional and phylogenetic diversity of ants (Arnan, Arcoverde, et al., 2018), and ant-plant interactions (Câmara et al., 2018; Oliveira et al., 2019, 2020). In a sense, aridity seems to impose a higher selective pressure than disturbance at different levels of biological organization. At the species-level, there are some species, such as *P. radoskowskii*, whose abundance does not respond greatly to increases in either aridity or disturbance (Oliveira et al., 2019), but that might adapt to new conditions through morphological adaptations (as we find here), which are strongly associated with physiology (Peters, 1983). Although *D. quadriceps* shows some morphological adaptations, it does not seem well adapted to increasing aridity as its populations declined along our aridity gradient (Oliveira et al., 2019). The other two focal species, *D. goeldi* and *E. muticum*, did not exhibit considerable changes in their abundances or morphology along the environmental gradients (Oliveira et al., 2019), but they can adapt behaviorally to increasing disturbance and aridity, shifting their foraging from sunny to shaded microhabitats, thus allowing their persistence in increasingly disturbed habitats (Silva et al., 2019). In fact, many ants use behavioral mechanisms to escape or avoid unfavorable conditions (Silva et al., 2019). Exploring the relevance of such mechanisms was beyond the scope of this study but is a potentially interesting avenue for future research on this topic.

In synthesis, the Caatinga dry forest supports a subset of ant fauna with high intraspecific morphological variability, which is in part associated with environmental gradients, particularly aridity, but also chronic anthropogenic disturbance. Such morphological variability is likely to represent a key asset by enabling individuals and populations to inhabit the immense variety of habitat provided by the Caatinga at the landscape and regional level. We refer to species with a high degree of generalist behavior relative to habitat use, food exploitation, and nesting (Hölldobler & Wilson, 1990; Leal et al., 2017), with potential impacts on community organization and structure, as well as the provision of ecosystem functions such as seed dispersal and plant protection against herbivores (Leal et al., 2017; Oliveira et al., 2019, 2020). This morphological variability may also represent a strategic asset for dealing with future changes in land use (i.e., land use intensification) and climate (increased aridity). Considering both the theoretical and applied implications of these findings, future studies should investigate the generality of high variability in Caatinga ant fauna, but also in the fauna of other dry forests, as all such biotas are experiencing changes in land use and climate.

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are openly available in the Dryad Digital Repository: <https://doi.org/10.5061/dryad.34tmpg4m9> (Oliveira et al., 2021).

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REFERENCES

- Andersen, A. N., & Majer, J. D. (2004). Ants show the way Down Under: Invertebrates as bioindicators in land management. *Frontiers in Ecology and the Environment*, 2, 291–298.
- Arnan, X., Arcoverde, G. B., Pie, M. R., Ribeiro-Neto, J. D., & Leal, I. R. (2018). Increased anthropogenic disturbance and aridity reduce phylogenetic and functional diversity of ant communities in Caatinga dry forest. *Science of the Total Environment*, 631–632, 429–438. <https://doi.org/10.1016/j.scitotenv.2018.03.037>
- Arnan, X., Cerdá, X., & Retana, J. (2012). Distinctive life traits and distribution along environmental gradients of dominant and subordinate Mediterranean ant species. *Oecologia*, 170, 489–500. <https://doi.org/10.1007/s00442-012-2315-y>
- Arnan, X., Leal, I. R., Tabarelli, M., Andrade, J. F., Barros, M. F., Câmara, T., Jamelli, D., Knoechelmann, C. M., Menezes, T. G. C., Menezes, A. G. S., Oliveira, F. M. P., de Paula, A. S., Pereira, S. C., Rito, K. F., Sfair, J. C., Siqueira, F. F. S., Souza, D. G., Specht, M. J., Vieira, L. A., ... Andersen, A. N. (2018). A framework for deriving measures of chronic anthropogenic disturbance: Surrogate, direct, single and multi-metric indices in Brazilian Caatinga. *Ecological Indicators*, 94, 274–282. <https://doi.org/10.1016/j.ecolind.2018.07.001>
- Barton, K. (2009). Mu-MIn: Multi-model inference. R Package Version 0.12.2/r18. <http://R-Forge.Rproject.org/projects/mumin/>
- Beever, E. A., Hall, L. E., Varner, J., Loosen, A. E., Dunham, J. B., Gahl, M. K., Smith, F. A., & Lawler, J. J. (2017). Behavioral flexibility as a mechanism for coping with climate change. *Frontiers in Ecology and the Environment*, 15, 299–308. <https://doi.org/10.1002/fee.1502>
- Bergmann, C. (1847). Über die verhältnisse der warmekonomie der-thiere zu ihrer grosse. *Gottinger Studien*, 1, 595–708.
- Bernadou, A., Romermann, C., Gratiashvili, N., & Heinze, J. (2016). Body size but not colony size increases with altitude in the holarctic ant, *Leptothorax acervorum*. *Ecological Entomology*, 41, 733–736.
- Bihn, J., Gebauer, G., & Brandl, R. (2010). Loss of functional diversity of ant assemblages in secondary tropical forests. *Ecology*, 91, 782–792. <https://doi.org/10.1890/08-1276.1>
- Bishop, T. R., Robertson, M. P., van Rensburg, B. J., & Parr, C. L. (2015). Contrasting species and functional beta diversity in montane ant assemblages. *Journal of Biogeography*, 42, 1776–1786. <https://doi.org/10.1111/jbi.12537>
- Blackburn, T. M., Gaston, K. J., & Loder, N. (1999). Geographic gradients in body size: A clarification of Bergmann's rule. *Diversity and Distributions*, 5, 165–174.
- Blois, J., Williams, J., Fitzpatrick, M., Jackson, S., & Ferrier, S. (2013). Space can substitute for time in predicting climate-change effects on biodiversity. *Proceedings of the National Academy of Sciences U S A*, 110, 9374–9379. <https://doi.org/10.1073/pnas.1220228110>
- Brown, W. L. (1953). Revisionary studies in the ant tribe Dacetini. *American Midland Naturalist*, 50, 1–137.
- Burnham, K. P., & Anderson, D. R. (2002). *Model selection and multimodel inference: A practical information theoretic approach*. Springer-Verlag.
- Caddy-Retalic, S., Andersen, A. N., Aspinwall, M. J., Breed, M. F., Byrne, M., Christmas, M. J., Dong, N., Evans, B. J., Fordham, D. A., Guerin, G. R., Hoffmann, A. A., Hughes, A. C., van Leeuwen, S. J., McInerney, F. A., Prober, S. M., Rossetto, M., Rymel, P. D., Steane, D. A., Wardle, G. M., & Lowe, A. J. (2017). Bioclimatic transect networks: Powerful observatories of ecological change. *Ecology and Evolution*, 7, 4607–4619. <https://doi.org/10.1002/ece3.2995>
- Câmara, T., Leal, I. R., Blüthgen, N., Oliveira, F. M. P., Queiroz, R. T., & Arnan, X. (2018). Effects of chronic anthropogenic disturbance and rainfall on the specialization of ant-plant mutualistic networks in the Caatinga, a Brazilian dry forest. *Journal of Animal Ecology*, 87, 1–12. <https://doi.org/10.1111/1365-2656.12820>
- Chown, S. L., & Nicolson, S. W. (2004). *Insect Physiological ecology: Mechanisms and patterns*. Oxford University Press.
- Davidson, D. W., Cook, S. C., & Snelling, R. R. (2004). Liquid-feeding performances of ants (Formicidae): Ecological and evolutionary implications. *Oecologia*, 139, 255–266. <https://doi.org/10.1007/s00442-004-1508-4>
- Fowler, H. G., Forti, L. C., Brandão, C. R., Delabie, J. H. C., & Vasconcelos, H. (1991). Ecologia nutricional de formigas. In A. R. Pannizzi & J. R. P. Parra (Eds.), *Ecologia nutricional de insetos* (pp. 141–223). Manole.
- Gaudard, C. A., Robertson, M. P., & Bishop, T. R. (2019). Low levels of intra-specific trait variation in a keystone invertebrate group. *Oecologia*, 190, 725–735. <https://doi.org/10.1007/s00442-019-04426-9>
- Gibb, H., & Parr, C. L. (2013). Does Structural Complexity Determine the Morphology of Assemblages? An Experimental Test on Three Continents. *PLoS One*, 8, e64005. <https://doi.org/10.1371/journal.pone.0064005>
- Gibb, H., Sanders, N., Dunn, R., Arnan, X., Vasconcelos, H., Donoso, D., Andersen, A. N., Silva, R., Bishop, T., Gómez, C., Grossman, B. F., Yusah, K. M., Luke, S., Pacheco, R., Pearce-Duvet, J., Retana, J., Tista, M., & Parr, C. (2018). Habitat disturbance selects against both small and large species across varying climates. *Ecography*, 41, 1184–1193. <https://doi.org/10.1111/ecog.03244>
- Gibb, H., Stoklosa, J., Warton, D. I., Brown, A. M., Andrew, N. R., & Cunningham, S. A. (2014). Does morphology predict trophic position and habitat use of ant species and assemblages? *Oecologia*, 177, 1–13. <https://doi.org/10.1007/s00442-014-3101-9>
- Griffiths, H. M., Louzada, J., Bardgett, R. D., & Barlow, J. (2016). Assessing the importance of intraspecific variability in dung beetle functional traits. *PLoS One*, 11, e0145598. <https://doi.org/10.1371/journal.pone.0145598>

- Hadfield Mg, S. M. (1996). Variability, flexibility and plasticity in life histories of marine invertebrates Gauthier-Villars (Ed.). *Oceanologica Acta*, 19, 323–334.
- Hölldobler, B., & Wilson, E. O. (1990). *The ants*. Harvard University Press.
- Jung, V., Violle, C., Mondy, C., Hoffmann, L., & Muller, S. (2010). Intraspecific variability and trait-based community assembly. *Journal of Ecology*, 98, 1134–1140. <https://doi.org/10.1111/j.1365-2745.2010.01687.x>
- Kaspari, M., & Weiser, M. D. (1999). The size–grain hypothesis and interspecific scaling in ants. *Functional Ecology*, 13, 530–538. <https://doi.org/10.1046/j.1365-2435.1999.00343.x>
- Kazakou, E., Violle, C., Roumet, C., Navas, M.-L., Vile, D., Kattge, J., & Garnier, E. (2014). Are trait-based species rankings consistent across data sets and spatial scales? *Journal of Vegetation Science*, 25, 235–247. <https://doi.org/10.1111/jvs.12066>
- Leal, I., Ribeiro Neto, J., Arnan, X., Oliveira, F., Arcoverde, G., Feitosa, R., & Andersen, A. (2017). Ants of the Caatinga: Diversity, biogeography, and functional responses to anthropogenic disturbance and climate change. In *Caatinga: The largest tropical dry forest region in South America*. pp. 65–95. Springer International Publishing.
- Legendre, P., & Legendre, L. (1998). *Numerical ecology* (2nd ed., p. 853). Elsevier.
- Maass, J. M. (1995). Conversion of tropical dry forest to pasture and agriculture. In E. Medina, H. A. Mooney, & S. H. Bullock (Eds.), *Seasonally dry tropical forests* (pp. 399–422). Cambridge University Press.
- Magrin, G., Marengo, J., Boulanger, J.-P., Buckeridge, M. S., Castellanos, E., Poveda, G., Scarano, F., & Vicuna, S. (2014). Central and South America. In: Climate change 2014: Impacts, Adaptation, and Vulnerability, Working Group II of the IPCC. pp. 1499–1566.
- Messier, J., McGill, B., & Lechowicz, M. (2010). How do traits vary across ecological scales? A case for trait-based ecology. *Ecology Letters*, 13, 838–848. <https://doi.org/10.1111/j.1461-0248.2010.01476.x>
- Moreau, C. S., & Bell, C. D. (2013). Testing the museum versus cradle tropical biological diversity hypothesis: phylogeny, diversification and ancestral biogeographic range evolution of the ants. *Evolution*, 67, 2240–2257.
- Okuzaki, Y., Takami, Y., & Sota, T. (2009). Resource partitioning or reproductive isolation: The ecological role of body size differences among closely related species in sympatry. *Journal of Animal Ecology*, 79, 383–392. <https://doi.org/10.1111/j.1365-2656.2009.01645.x>
- Oliveira, F., Andersen, A., Arnan, X., Ribeiro Neto, J., Arcoverde, G., & Leal, I. (2019). Effects of increasing aridity and chronic anthropogenic disturbance on seed dispersal by ants in Brazilian Caatinga. *Journal of Animal Ecology*, 88, 870–880. <https://doi.org/10.1111/1365-2656.12979>
- Oliveira, F. M. P., Câmara, T., Durval, J. I. F., Oliveira, C. L. S., Arnan, X., Andersen, A. N., Ribeiro, E. M. S., & Leal, I. R. (2020). Plant protection services mediated by extrafloral nectaries decline with aridity but are not influenced by chronic anthropogenic disturbance in Brazilian Caatinga. *Journal of Ecology*, 00, 1–13. <https://doi.org/10.1111/1365-2745.13469>
- Oliveira, F. M. P., Ribeiro-Neto, J. D., Andersen, A. N., & Leal, I. R. (2017). Chronic anthropogenic disturbance as a secondary driver of ant community structure: interactions with soil type in Brazilian Caatinga. *Environmental Conservation*, 44, 115–123. <https://doi.org/10.1017/S0376892916000291>
- Oliveira, F. M. P., Silva, L. L., Leal, I. R., & Arnan, X. (2021). Data from: Morphology of four common and phylogenetically distant ant species varies along disturbance and aridity gradients in the Caatinga dry forest. *Dryad Digital Repository*. <https://doi.org/10.5061/dryad.344tmpg4m9>
- Oms, C. S., Cerdá, X., & Boulay, R. (2017). Is phenotypic plasticity a key mechanism for responding to thermal stress in ants? *The Science of Nature*, 104, 42. <https://doi.org/10.1007/s00114-017-1464-6>
- Parr, C. L. et al (2017). GlobalAnts: A new database on the geography of ant traits (Hymenoptera: Formicidae). *Insect Conservation and Diversity*, 10, 5–20.
- Peters, R. H. (1983). *The ecological implications of body size*. Cambridge University Press.
- Pigot, A. L., Trisos, C. H., & Tobias, J. A. (2016). Functional traits reveal the expansion and packing of ecological niche space underlying an elevational diversity gradient in passerine birds. *Proceedings of the Royal Society B-Biological Sciences*, 283, 20152013. <https://doi.org/10.1098/rspb.2015.2013>
- Purcell, J., Pirogan, D., Avril, A., Bouyarden, F., & Chapuisat, M. (2016). Environmental influence on the phenotype of ant workers revealed by common garden experiment. *Behavioral Ecology and Sociobiology*, 70, 357–367. <https://doi.org/10.1007/s00265-015-2055-1>
- R Core Team (2016). *R: A language and environment for statistical Computing*. R Foundation for Statistical Computing, Vienna. <http://www.R-project.org>
- Rito, K. F., Arroyo-Rodríguez, V., Queiroz, R. T., Leal, I. R., & Tabarelli, M. (2017). Precipitation mediates the effect of human disturbance on the Brazilian Caatinga vegetation. *Journal of Ecology*, 105, 828–838. <https://doi.org/10.1111/1365-2745.12712>
- Sanaphre-Villanueva, L., Dupuy, J., Andrade, J., Reyes, C., Paz, H., & Jackson, P. (2016). Functional diversity of small and large trees along secondary succession in a tropical dry forest. *Forests*, 7, 163. <https://doi.org/10.3390/f7080163>
- Sfair, J. C., de Bello, F., de França, T. Q., Baldauf, C., & Tabarelli, M. (2018). Chronic human disturbance affects plant trait distribution in a seasonally dry tropical forest. *Environmental Research Letters*, 13, 25005. <https://doi.org/10.1088/1748-9326/aa9f5e>
- Shik, J., Arnan, X., Oms, C., Cerdá, X., & Boulay, R. (2019). Evidence for locally adaptive metabolic rates among ant populations along an elevational gradient. *Journal of Animal Ecology*, 88, 1240–1249. <https://doi.org/10.1111/1365-2656.13007>
- Silva, I., Leal, I., Ribeiro Neto, J., & Arnan, X. (2019). Spatiotemporal responses of ant communities across a disturbance gradient: the role of behavioral traits. *Insectes Sociaux*, 66, 623–635. <https://doi.org/10.1007/s00040-019-00717-9>
- Silva, J. M., Leal, I., & Tabarelli, M. (2017). *Caatinga: The largest tropical dry forest region in South America*. Springer International Publishing.
- Silva, R. R., & Brandão, C. R. F. (2010). Morphological patterns and community organization in leaf-litter ant assemblages. *Ecological Monographs*, 80, 107–124. <https://doi.org/10.1890/08-1298.1>
- Southwood, T. R. E. (1988). Tactics, strategies and templates. *Oikos*, 52, 3–18.
- Stouffer, D., Rezende, E., & Amaral, L. (2011). The role of body mass in diet contiguity and food-web structure. *Journal of Animal Ecology*, 80, 632–639. <https://doi.org/10.1111/j.1365-2656.2011.01812.x>
- Trabucco, A., & Zomer, R. J. (2010). Global Soil Water Balance Geospatial Database CGIAR Consortium for Spatial Information. <http://www.cgiar-csi.org>
- Underwood, E. C., & Fisher, B. L. (2006). The role of ants in conservation monitoring: If, when, and how. *Biological Conservation*, 132, 166–182. <https://doi.org/10.1016/j.biocon.2006.03.022>
- Violle, C., Navas, M.-L., Vile, D., Kazakou, E., Fortunel, C., Hummel, I., & Garnier, E. (2007). Let the concept of trait be functional! *Oikos*, 116, 882–892. <https://doi.org/10.1111/j.0030-1299.2007.15559.x>
- Weeks, B. C., Willard, D. E., Zimova, M., Ellis, A. A., Witynski, M. L., Hennen, M., & Winger, B. M. (2020). Shared morphological consequences of global warming in North American migratory birds. *Ecology Letters*, 23, 316–325. <https://doi.org/10.1111/ele.13434>
- Weiser, M. D., & Kaspari, M. (2006). Ecological morphospace of New World ants. *Ecological Entomology*, 31, 131–142. <https://doi.org/10.1111/j.0307-6946.2006.00759.x>
- Wittlinger, M., Wehner, R., & Wolf, H. (2007). The desert ant odometer: A stride integrator that accounts for stride length and walking

speed. *Journal of Experimental Biology*, 210, 198–207. <https://doi.org/10.1242/jeb.02657>

Yates, M., Andrew, N., Binns, M., & Gibb, H. (2014). Morphological traits: Predictable responses to macrohabitats across a 300 km scale. *PeerJ*, 2, e271. <https://doi.org/10.7717/peerj.271>

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